

the upside is that it can be run frequently and quickly, often in a matter of minutes (rather than hours or days). This allows for a much quicker turnaround time on potentially time-consuming and critical issues.

Mimic Production

Each new deployment stage should be made into an environment that mimics the actual final production environment as closely as possible. This includes infrastructure, operating system, databases, patches, network, firewalls and configuration.

By validating new software changes in this kind of realistic pre-production environment, mismatches and last-minute surprises can be effectively eliminated and applications can be released safely to production with a higher assurance of success.

Dont forget the Database

Continuous delivery best practices would never be complete without advising that the database be subjected to the same basic quality protocols that ensure secure and reliable source code, task, configuration, build and deployment management.

However, many times this is neglected because the database is rather difficult to manage or incrementally grows by doing something as easy as copying files from one location to another. Yet, the database is actually a repository of the most valued and irreplaceable asset — our data — and preserving it accurately is imperative to continuous delivery.

Although the database brings with it several unique challenges, specialized database tools such as enforced database source control (for all environments), a database build automation tool along with database release and verification processes can ensure a stable resource in your DevOps chain.

Considering these continuous delivery best practices will allow you to create a deployment pipeline with increased productivity, faster time to market, reduced risk and increased quality.

Like all other stages mentioned before, deployment is an important part of running an application but the unique position of this stage (in effect the last one) also ensures that all the deliverables generated in the previous stages would only be delivered if the deployment stage completes successfully. Having a robust, fault tolerant and rapid deployment process is therefore paramount to achieve our final goal in software development: To get the application running in the production environment. **d**